Δ DELTA FILINGS

Corporate Governance Intelligence Platform

Business Requirements Document

Version 2.0 — April 2026

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Document Control

Revision History

Version	Date	Author	Summary of Changes
0.1	2025-09-12	Product Team	Initial draft — scope and objectives.
0.5	2025-12-04	Product Team	Added functional and non-functional requirements.
1.0	2026-02-18	Product Team	Reviewed and approved for v1 release.
2.0	2026-04-23	Product Team	v2 scope: Postgres + Redis + containerised stack, director compare, compliance module.

Approvers

Name	Role	Organisation	Signed On
Prakhar Singh	Product Owner	Delta Filings	2026-04-23
Engineering Lead	Technical Sign-off	Delta Filings	2026-04-23
Compliance Advisor	Domain Sign-off	Delta Filings	2026-04-23

Distribution List

Recipient	Role	Access Level
Executive Sponsor	Strategic oversight	Full
Engineering Team	Implementation	Full
QA Team	Validation	Full
External Auditor	Compliance review	Read-only

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1. Executive Summary

Delta Filings is a corporate governance intelligence platform that gives investors, compliance officers, analysts and researchers a single source of truth for filings, director records and compliance posture of Indian listed companies. The platform ingests data from statutory sources (BSE, NSE, MCA) and surfaces it through a modern web experience backed by a structured database and AI-driven summaries.

This Business Requirements Document (BRD) formalises the scope, objectives and requirements of Delta Filings v2 — a re-architected release that moves from a single-process Flask application to a containerised, multi-service platform built on FastAPI, Next.js, PostgreSQL and Redis. The v2 scope also introduces a dedicated Compliance module, a Director Compare workflow and a Watchlist-driven alerting layer.

The document is intended to align product, engineering, compliance and leadership stakeholders on what will be built, why it is being built and the criteria by which success will be measured.

2. Business Context and Problem Statement

2.1 The Problem

Indian listed companies generate thousands of statutory disclosures every week across BSE, NSE and MCA. These disclosures are distributed across three portals, served as inconsistent PDFs, and lack any structured way to cross-reference directorships, historical filings or compliance events. As a result, investors and compliance teams spend disproportionate time on manual collection rather than analysis.

2.2 Who Is Affected

- Retail and institutional investors tracking corporate events.
- Compliance officers monitoring obligations under LODR, Companies Act and SEBI regulations.
- Equity research analysts preparing coverage notes and governance reports.
- Legal and audit teams conducting due-diligence on companies and directors.

2.3 Current State

Delta Filings v1 (live since 2025) demonstrated strong product-market fit for filing aggregation and DIN lookup. However, it was built as a single Flask process with a SQLite backing store, making horizontal scaling, caching and background ingestion difficult. v1 also lacked a first-class compliance workflow and director comparison, both of which were the most-requested features from early users.

2.4 Opportunity

A structured, real-time governance intelligence layer for Indian capital markets is an under-served category. Delta Filings v2 will convert the prototype learnings into a production-grade platform that can be extended into regulated verticals (audit, legal, banking) and white-labeled for enterprise customers.

3. Business Objectives and Success Metrics

3.1 Strategic Objectives

1. Establish Delta Filings as the default research surface for Indian corporate filings.
2. Reduce analyst time-to-insight on any listed company from hours to minutes.
3. Build a defensible data layer (filings + directors + compliance) that compounds over time.
4. Create an enterprise-ready architecture capable of supporting paid tiers within 12 months.

3.2 Key Performance Indicators

KPI	Definition	v2 Target	Cadence
Monthly Active Users	Unique signed-in users / month	5,000	Monthly
Filings Coverage	% of BSE/NSE top-500 filings captured within 15 min	≥ 98%	Weekly
DIN Lookup Latency (p95)	Server time for a director lookup	< 800 ms	Continuous
Compliance Run Success	Completed runs / total runs	≥ 95%	Weekly
User Retention (Week 4)	% of new users active in week 4	≥ 35%	Monthly
Paid Conversion	Free to paid conversion, quarter	≥ 4%	Quarterly

4. Scope

4.1 In Scope (v2)

- Real-time ingestion of BSE and NSE corporate announcements.
- MCA director record scraping with cached DIN lookup.

- User watchlists with price and filing alerts.
- AI-generated filing summaries and materiality classification.
- Director Compare — side-by-side view of up to three directors.
- Compliance module — obligations dashboard, run history, audit trail.
- Role-based authentication via Supabase (email and Google).
- Containerised deployment (Docker Compose for dev, orchestrated for prod).
- PostgreSQL as system of record; Redis as cache and job queue.

4.2 Out of Scope (v2)

- Retail trading, order placement or brokerage integrations.
- Personalised investment or tax advisory.
- Coverage of unlisted private companies beyond MCA director data.
- Non-Indian capital markets.
- Mobile native applications (web-responsive only in v2).

4.3 Assumptions

- BSE, NSE and MCA continue to expose their current public data channels.
- Supabase remains the identity provider for v2.
- OpenAI API availability and pricing remain within budgeted limits.
- Users have modern evergreen browsers (Chrome, Edge, Safari, Firefox — latest two versions).

4.4 Constraints

- No storage of card numbers, bank credentials or sensitive personal identifiers.
- All filings content must be attributed to its original source URL.
- Infrastructure must be reproducible via Docker Compose for local development.
- Initial release must comply with Indian data-localisation expectations for user data.

5. Stakeholders

The table below captures the stakeholders involved in Delta Filings v2, their primary interests and the nature of their engagement.

Stakeholder	Interest	Engagement
Product Owner	Overall outcome, prioritisation, release decisions	Decision-maker

Stakeholder	Interest	Engagement
Engineering Lead	Architecture, delivery, quality	Decision-maker
Compliance Advisor	Accuracy of regulatory content	Reviewer
Design Lead	Usability and visual system	Contributor
QA Lead	Functional and regression validation	Reviewer
Pilot Customers	Beta usage and feedback	Contributor
Executive Sponsor	Strategic fit and investment	Informed

6. User Personas

6.1 Arjun — Equity Research Analyst

- Covers 40+ mid-cap stocks across BFSI and pharma.
- Needs: filing summaries within 10 minutes of announcement, director background in one click.
- Pain today: juggles BSE, NSE and MCA tabs; loses time reconciling data.

6.2 Meera — Compliance Officer

- Accountable for quarterly LODR and Companies Act filings for a listed company.
- Needs: obligation tracker, evidence capture, audit-ready export.
- Pain today: spreadsheets, email threads, no single source of truth.

6.3 Raghav — Retail Investor / Finfluencer

- Manages a personal portfolio and publishes weekly newsletters.
- Needs: alerts when tracked companies file something material.
- Pain today: information overload; struggles to separate signal from noise.

6.4 Priya — Legal and Audit Professional

- Performs due-diligence on directors and companies.
- Needs: historical director associations, cross-company directorships, compliance history.
- Pain today: MCA portal usability, poor search, no cross-company view.

7. Business Requirements

Business requirements describe what the business needs the platform to do, irrespective of implementation. Each requirement has a unique identifier (BR-###) that maps forward to the Functional and Non-Functional Requirements.

7.1 Filings and Announcements

ID	Requirement	Priority
BR-001	Users can search filings by company name, ticker or ISIN.	Must
BR-002	Users can filter filings by date, category and exchange.	Must
BR-003	Users receive AI-generated summaries of each filing.	Must
BR-004	Users can save filings to a personal watchlist.	Should
BR-005	Users can export filings metadata to CSV.	Should

7.2 Directors and DIN Lookup

ID	Requirement	Priority
BR-010	Users can look up any director by DIN.	Must
BR-011	The platform shows all companies a director is associated with.	Must
BR-012	Users can compare up to three directors side-by-side.	Must
BR-013	Directors flagged as disqualified are visually distinguished.	Should
BR-014	Users can bookmark directors for monitoring.	Could

7.3 Compliance Module

ID	Requirement	Priority
BR-020	Compliance officers can list obligations applicable to a company.	Must
BR-021	Users can trigger a compliance run and view its status.	Must
BR-022	Each run records evidence, timestamp and outcome for audit.	Must
BR-023	The module supports export of a compliance report (PDF).	Should

ID	Requirement	Priority
BR-024	Users receive reminders ahead of statutory deadlines.	Should

7.4 Watchlists and Alerts

ID	Requirement	Priority
BR-030	Users can create and name multiple watchlists.	Must
BR-031	Users receive email alerts on new filings in a watchlist.	Must
BR-032	Alert frequency is configurable (instant, daily digest, weekly).	Should
BR-033	Users can mute alerts for specific filing categories.	Could

7.5 Accounts, Access and Billing

ID	Requirement	Priority
BR-040	Users can register via email or Google OAuth.	Must
BR-041	Users can manage their profile and notification preferences.	Must
BR-042	The platform supports Free, Pro and Enterprise tiers.	Should
BR-043	Enterprise tier supports SSO and team seats.	Could

8. Functional Requirements

Functional requirements translate the business requirements above into capabilities of the system. They are grouped by module. Priority uses the MoSCoW convention (Must, Should, Could, Won't in this release).

8.1 Dashboard Module

- FR-D-01 — The dashboard shall display the latest 50 filings across the user's watchlists.
- FR-D-02 — The dashboard shall display a sentiment-tagged AI summary for each filing.
- FR-D-03 — The dashboard shall display the health status of the platform's ingestion pipelines.

8.2 Announcements Module

- FR-A-01 — The system shall fetch new filings from BSE and NSE every 60 seconds during market hours.

- FR-A-02 — The system shall de-duplicate identical filings cross-listed on both exchanges.
- FR-A-03 — The system shall classify each filing into a materiality tier (High, Medium, Low).
- FR-A-04 — The system shall expose a filing-detail page with source PDF, summary and related filings.

8.3 DIN Search Module

- FR-N-01 — The system shall accept an 8-digit DIN and return the director's name, DOB masked to year, and associations.
- FR-N-02 — The system shall cache director lookups in Redis for 24 hours.
- FR-N-03 — The system shall show a distinct 'curated snapshot' badge when data is served from the curated set rather than live MCA.

8.4 Compare Module

- FR-C-01 — The user shall be able to add up to three directors to a comparison view.
- FR-C-02 — The comparison view shall highlight common directorships.
- FR-C-03 — The comparison view shall export to a shareable link (read-only).

8.5 Compliance Module

- FR-M-01 — The system shall present obligations relevant to the selected company based on its listing and sector.
- FR-M-02 — The user shall be able to create a compliance run with a due date.
- FR-M-03 — The system shall capture each evidence attachment against its obligation.
- FR-M-04 — The user shall export a run summary as a PDF suitable for audit review.

8.6 Watchlist and Alerts

- FR-W-01 — The user shall create, rename, and delete watchlists.
- FR-W-02 — The system shall send email alerts via a transactional provider.
- FR-W-03 — The user shall receive a weekly digest by default with an opt-out link.

8.7 Account Management

- FR-U-01 — Authentication is delegated to Supabase using email and Google OAuth.
- FR-U-02 — Profiles store display name, organisation and notification preferences.
- FR-U-03 — The system shall honour account deletion requests within 30 days.

9. Non-Functional Requirements

9.1 Performance

Metric	Target
Dashboard time-to-interactive (p95)	< 2.5 seconds on 4G
API latency (p95, authenticated endpoints)	< 400 ms
Filings ingestion lag from exchange publication	< 15 minutes
Concurrent user support (v2 launch)	2,000

9.2 Availability and Reliability

- Platform uptime target: 99.5% monthly during market hours.
- Planned maintenance windows outside market hours, announced 48 hours in advance.
- Automated daily backups of the PostgreSQL database, retained for 30 days.

9.3 Security

- TLS 1.2+ for all external traffic; HSTS enabled.
- All secrets managed via environment variables; no secrets committed to source control.
- JWT-based session tokens with 60-minute expiry and refresh rotation.
- Principle of least privilege for database and service accounts.
- Quarterly dependency vulnerability review (Dependabot, OSV).

9.4 Privacy and Data Protection

- Personally identifying information limited to email, display name and organisation.
- No storage of card numbers, bank credentials or government IDs.
- Data-subject access and deletion requests honoured within 30 days.
- User data residency: Indian region for primary storage, subject to provider availability.

9.5 Accessibility

- WCAG 2.1 AA as the baseline for all customer-facing surfaces.
- Keyboard-only navigation across all primary flows.
- Colour contrast ratios validated at design review.

9.6 Observability

- Structured JSON logs shipped to a central log aggregator.
- Per-endpoint latency and error-rate dashboards.
- Synthetic probes for dashboard and DIN lookup every 60 seconds.
- On-call rotation with 15-minute acknowledgement SLO for Sev-1 incidents.

10. Solution Overview

Delta Filings v2 is a containerised multi-service application. The table below summarises the principal components. A more detailed technical design is maintained separately in the architecture document.

Component	Technology	Purpose
Frontend	Next.js 15 (App Router), React 18, Tailwind CSS	Server-rendered user interface and client interactions.
Backend API	FastAPI (Python 3.12), SQLAlchemy 2 async	Business logic and REST endpoints.
Database	PostgreSQL 16	System of record for users, filings, compliance runs.
Cache and Queue	Redis 7	Caching, dedup, background job queue.
Authentication	Supabase	Email and Google OAuth, JWT issuance.
AI Summaries	OpenAI API	Filing summarisation and materiality classification.
Ingestion Workers	Python async scrapers (BSE, NSE, MCA)	Fetch, normalise and persist filings.
Orchestration	Docker Compose (dev), container platform (prod)	Reproducible deployment.

11. Data Requirements

11.1 Primary Entities

- Company — identified by symbol, ISIN and CIN.
- Filing — timestamped disclosure with source URL and category.
- Director — identified by DIN, with masked DOB.

- DirectorAssociation — link between a director and a company with role and tenure.
- User Profile — email, display name, organisation, preferences.
- Watchlist — user-owned collection of companies.
- Compliance Run — instance of an obligation with status and evidence.

11.2 Data Sources

- BSE corporate announcements feed.
- NSE corporate announcements feed.
- MCA public director records (scraped, rate-limited).
- Curated director set maintained by Delta Filings editorial.

11.3 Data Quality Targets

- Filing ingestion completeness: ≥ 98% of BSE/NSE top-500 filings captured.
- Filing categorisation accuracy: ≥ 95% against a weekly spot-check sample.
- Director association freshness: ≤ 7 days since last reconciliation.

11.4 Data Retention

- Filings: retained indefinitely.
- Audit logs: retained 18 months.
- User activity telemetry (non-PII): retained 12 months.
- Deleted user data: purged within 30 days of request.

12. Regulatory and Compliance Considerations

- All filings content must link back to the originating BSE/NSE/MCA URL.
- The platform shall not claim to replace statutory filing channels.
- AI-generated summaries shall carry a visible 'machine-generated' label.
- User data handling follows the expectations of the Digital Personal Data Protection Act, 2023.
- Terms of Service and Privacy Policy shall be reviewed legally before v2 launch.

13. Risks and Mitigations

ID	Risk	Impact	Likelihood	Mitigation
R-01	Upstream exchange portal	High	Medium	Source abstraction layer; synthetic

ID	Risk	Impact	Likelihood	Mitigation
	changes break scrapers.			probes; 24-hour fix SLO.
R-02	AI summary hallucination damages credibility.	High	Low	Confidence threshold; 'machine-generated' label; editorial spot-checks.
R-03	MCA rate-limiting blocks director lookups.	Medium	Medium	Aggressive Redis caching; curated snapshot fallback.
R-04	Database growth outpaces plan.	Medium	Low	Partitioning on filings by year; archival tier for >2-year-old PDFs.
R-05	Supabase outage prevents login.	High	Low	Graceful degradation; public read surfaces remain available.
R-06	Key team member attrition.	Medium	Medium	Documentation-first culture; rotation of on-call duties.

14. Assumptions and Dependencies

14.1 External Dependencies

- Supabase (identity and user management).
- OpenAI (summarisation and classification).
- Transactional email provider (alerts and digests).
- BSE, NSE, MCA public data channels.

14.2 Internal Dependencies

- Design system is complete before front-end implementation of each module.
- Legal review of Terms of Service and Privacy Policy before v2 launch.
- Editorial pipeline for curated director records staffed ahead of launch.

15. Release Plan and Roadmap

Milestone	Target Date	Scope
M1 — Platform scaffold	2026-05-15	Containerised stack, authentication, empty dashboard.

Milestone	Target Date	Scope
M2 — Filings live	2026-06-15	Live BSE/NSE ingestion, filing detail page, AI summaries.
M3 — Directors and DIN	2026-07-10	DIN lookup, director compare, curated snapshot.
M4 — Compliance module	2026-08-15	Obligations catalogue, runs, evidence, PDF export.
M5 — Watchlists and alerts	2026-09-15	Email alerts, digest cadence, mute controls.
M6 — Public launch	2026-10-01	Free tier launch, onboarding, documentation site.
M7 — Paid tiers	2026-12-01	Pro and Enterprise billing, SSO for Enterprise.

16. Acceptance Criteria

The release is considered acceptable when all of the following are demonstrable on the staging environment for two consecutive weeks without regression:

5. All 'Must' priority functional requirements pass their acceptance tests.
6. Performance targets in Section 9.1 are met in load testing at 2x launch concurrency.
7. All 'High' severity security findings are remediated.
8. Accessibility audit produces no Level A or AA blocking issues.
9. On-call runbooks cover ingestion, auth, AI and database failure modes.
10. A customer dry-run with three pilot users completes end-to-end workflows without facilitator intervention.

Appendix A — Glossary

Term	Definition
BRD	Business Requirements Document — this document.
BSE	Bombay Stock Exchange.
CIN	Corporate Identification Number issued by MCA.
DIN	Director Identification Number issued by MCA.

Term	Definition
DPDP Act	Digital Personal Data Protection Act, 2023 (India).
ISIN	International Securities Identification Number.
LODR	SEBI (Listing Obligations and Disclosure Requirements) Regulations, 2015.
MCA	Ministry of Corporate Affairs, Government of India.
MoSCoW	Prioritisation convention — Must, Should, Could, Won't.
NSE	National Stock Exchange of India.
SLO	Service Level Objective — target level for a reliability metric.
SSO	Single Sign-On.
TLS	Transport Layer Security, the cryptographic protocol for HTTPS.
WCAG	Web Content Accessibility Guidelines.